KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5331 | 5 Credits: 4 | Program core | Source-grounded

Industrial Automation Control

 To introduce the concepts of Process and Process control.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5331 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5331.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Automation and Robotics
Course code	5331
Course title	Industrial Automation Control
Semester	5 Credits: 4
Course category	Program core
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

17 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To introduce the concepts of Process and Process control.
- To illustrates various elements in a process control loop and controller tuning.
- Familiarize the basic concepts of PLC programming
- To explain functional elements of process control loop and various
- communication methods in process control applications.

Course outcomes

CO	Official outcome	Study level
CO1	Understand the concept of Process control and the 15 Understanding process control loops	See official syllabus
CO2	To understand the various control modes and to 16 Applying implement pneumatic and electronic controllers	See official syllabus
CO3	Familiarize the basic concepts of PLC programming 16 Understanding To understand telemetry system and Digital	See official syllabus
CO4	13 Applying Communication channels in Process Control	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3 2
CO3	CO3 3

Row	Extracted official mapping
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Understand the concept of Process control and the process control loops	4	As specified
Module 2	and electronic controllers	4	As specified
Module 3	Familiarize basic concept of PLC programming	6	As specified
Module 4	Process Control	3	As specified

MODULE 1

Understand the concept of Process control and the process control loops

This module is presented from the official syllabus scope for Industrial Automation Control. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.

- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Understand the concept of Process control and the process control loops
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Define the concepts of process control.

Define the concepts of process control. is an official topic in Module 1 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define the concepts of process control. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define the concepts of process control. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define the concepts of process control. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Define the concepts of process control. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain automatic control 3 Understanding Explain the process and controller

Explain automatic control 3 Understanding Explain the process and controller is an official topic in Module 1 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain automatic control 3 Understanding Explain the process and controller using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain automatic control 3 understanding explain the process and controller should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain automatic control 3 Understanding Explain the process and controller means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain automatic control 3 Understanding Explain the process and controller □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

4 Understanding Characteristics

4 Understanding Characteristics is an official topic in Module 1 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding Characteristics using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding characteristics should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding Characteristics means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 4 Understanding Characteristics [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Describe the different process control loops 5 Understanding Automatic control-Block diagram of process control - Error, set point, controlled variable, manipulated variable and measured variable-Elements of process control loop-Process characteristics - Process equation, Process Load, Process Lag and Self regulation, Control System Parameters - Error, Variable Range, Control Parameter Range, Control Lag –Dead Time, Cycling-Process Parameters-Temperature process control, Pressure Process control system-Flow process control system, Level process control system. To understand the various control modes and to implement pneumatic

Describe the different process control loops 5 Understanding Automatic control-Block diagram of process control - Error, set point, controlled variable, manipulated variable and measured variable-Elements of process control loop-Process characteristics - Process equation, Process Load, Process Lag and Self regulation, Control System Parameters - Error, Variable Range, Control Parameter Range, Control Lag –Dead Time, Cycling-Process Parameters-Temperature process control, Pressure Process control system-Flow process control system, Level process control system. To understand the various control modes and to implement pneumatic is an official topic in Module 1 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the different process control loops 5 Understanding Automatic control-Block diagram of process control - Error, set point, controlled variable, manipulated variable and measured variable-Elements of process control loop-Process characteristics - Process equation, Process Load, Process Lag and Self regulation, Control System Parameters - Error, Variable Range, Control Parameter Range, Control Lag –Dead Time, Cycling-Process Parameters-Temperature process control, Pressure Process control system-Flow process control system, Level process control system. To understand the various control modes and to implement pneumatic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the different process control loops 5 understanding automatic control-block diagram of process control - error, set point, controlled variable, manipulated variable and measured variable-elements of process control loop-process characteristics - process equation, process load, process lag and self regulation, control system parameters - error, variable range, control parameter range, control lag –dead time, cycling-process parameters-temperature process control, pressure process control system-flow process control system, level process control system. to understand the various control modes and to implement pneumatic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the different process control loops 5 Understanding Automatic control-Block diagram of process control - Error, set point, controlled variable, manipulated variable and measured variable-Elements of process control loop-Process characteristics - Process equation, Process Load, Process Lag and Self regulation, Control System Parameters - Error, Variable Range, Control Parameter Range, Control Lag –Dead Time, Cycling-Process Parameters-Temperature process control, Pressure Process control system-Flow process control system, Level process control system. To understand the various control modes and to implement pneumatic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Describe the different process control loops 5 Understanding Automatic control-Block diagram of process control - Error, set point, controlled variable, manipulated variable and measured variable-Elements of process control loop-Process characteristics - Process equation, Process Load, Process Lag and Self regulation, Control System Parameters - Error, Variable Range, Control Parameter Range, Control Lag –Dead Time, Cycling-Process Parameters-Temperature process control, Pressure Process control system-Flow process control system, Level process control system. To understand the various control modes and to implement pneumatic □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Understand the concept of Process control and the process control loops

  M1.01      Define the concepts of process control.               3
Understanding

  M1.02      Explain automatic control                             3
Understanding

             Explain the process and controller
  M1.03                                                            4
Understanding
             Characteristics

  M1.04      Describe the different process control loops          5
Understanding

Contents: Process- Process Control and Process Plant -Human aided control and
Automatic control-Block diagram of process control - Error, set point, controlled
variable,
manipulated variable and measured variable-Elements of process control loop-
Process
characteristics - Process equation, Process Load, Process Lag and Self
regulation, Control
System Parameters - Error, Variable Range, Control Parameter Range, Control Lag –
Dead
Time, Cycling-Process Parameters-Temperature process control, Pressure Process
control
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

and electronic controllers

This module is presented from the official syllabus scope for Industrial Automation Control. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: and electronic controllers
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Describe different types of process 4 Understanding To understand the various control modes

Describe different types of process 4 Understanding To understand the various control modes is an official topic in Module 2 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe different types of process 4 Understanding To understand the various control modes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe different types of process 4 understanding to understand the various control modes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe different types of process 4 Understanding To understand the various control modes means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Describe different types of process 4 Understanding To understand the various control modes □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

6 Understanding continuous and discontinuous Understand the implementation of Electronic

6 Understanding continuous and discontinuous Understand the implementation of Electronic is an official topic in Module 2 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding continuous and discontinuous Understand the implementation of Electronic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding continuous and discontinuous understand the implementation of electronic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding continuous and discontinuous Understand the implementation of Electronic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 6 Understanding continuous and discontinuous Understand the implementation of Electronic □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

3 Understanding Controllers

3 Understanding Controllers is an official topic in Module 2 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Controllers using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding controllers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Controllers means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 3 Understanding Controllers [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Illustrate Tuning of controllers 3 Applying neutralzone -multi position control Mode- Continuous Control Modes - proportional controlMode -proportional band and offset error - integral control Mode – reset rate – derivative control Mode – derivative time- composite control Modes- PI, PD and PID control Modes–Electronic Controllers-Electronic Error detector-Electronic P,PI,PD and PID controller using Op -amp-Tuning methods– open loop method-Ziegler Nichols method.

Illustrate Tuning of controllers 3 Applying neutralzone -multi position control Mode- Continuous Control Modes - proportional controlMode -proportional band and offset error - integral control Mode – reset rate – derivative control Mode – derivative time- composite control Modes- PI, PD and PID control Modes–Electronic Controllers-Electronic Error detector-Electronic P,PI,PD and PID controller using Op -amp-Tuning methods– open loop method-Ziegler Nichols method. is an official topic in Module 2 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate Tuning of controllers 3 Applying neutralzone -multi position control Mode- Continuous Control Modes - proportional controlMode -proportional band and offset error - integral control Mode – reset rate – derivative control Mode – derivative time- composite control Modes- PI, PD and PID control Modes–Electronic Controllers-Electronic Error detector-Electronic P,PI,PD and PID controller using Op -amp-Tuning methods– open loop method-Ziegler Nichols method. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate tuning of controllers 3 applying neutralzone -multi position control mode- continuous control modes - proportional controlmode -

proportional band and offset error - integral control mode – reset rate – derivative control mode – derivative time- composite control modes- pi, pd and pid control modes–electronic controllers-electronic error detector-electronic p,pi,pd and pid controller using op -amp-tuning methods– open loop method-ziegler nichols method. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate Tuning of controllers 3 Applying neutralzone -multi position control Mode- Continuous Control Modes - proportional controlMode -proportional band and offset error - integral control Mode – reset rate – derivative control Mode – derivative time- composite control Modes- PI, PD and PID control Modes–Electronic Controllers-Electronic Error detector-Electronic P,PI,PD and PID controller using Op -amp-Tuning methods– open loop method-Ziegler Nichols method. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Illustrate Tuning of controllers 3 Applying neutralzone -multi position control Mode- Continuous Control Modes - proportional controlMode -proportional band and offset error - integral control Mode – reset rate – derivative control Mode – derivative time- composite control Modes- PI, PD and PID control Modes–Electronic Controllers-Electronic Error detector-Electronic P,PI,PD and PID controller using Op -amp-Tuning methods– open loop method-Ziegler Nichols method. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
and electronic controllers

  M2.01      Describe different types of process                   4
Understanding


             To understand the various control modes
  M2.02                                                            6
Understanding
             continuous and discontinuous
            Understand the implementation of Electronic
  M2.03                                                         3
Understanding
            Controllers

  M2.04     Illustrate Tuning of controllers                    3
Applying


            Series Test – I                                     1

Contents: Batch and Continuous process - Discontinuous control Modes - two
position -
neutralzone -multi position control Mode- Continuous Control Modes - proportional
controlMode -proportional band and offset error - integral control Mode – reset
rate –
derivative control Mode – derivative time- composite control Modes- PI, PD and
PID
control Modes–Electronic Controllers-Electronic Error detector-Electronic P,PI,PD
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Familiarize basic concept of PLC programming

This module is presented from the official syllabus scope for Industrial Automation Control. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Familiarize basic concept of PLC programming
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

Understand Sequence and logic control 1 Understanding Describe a programmable logic controller and

Understand Sequence and logic control 1 Understanding Describe a programmable logic controller and is an official topic in Module 3 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand Sequence and logic control 1 Understanding Describe a programmable logic controller and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand sequence and logic control 1 understanding describe a programmable logic controller and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand Sequence and logic control 1 Understanding Describe a programmable logic controller and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Understand Sequence and logic control 1 Understanding Describe a programmable logic controller and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

2 Understanding its components

2 Understanding its components is an official topic in Module 3 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding its components using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding its components should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding its components means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 2 Understanding its components [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Understand Ladder diagram

Understand Ladder diagram is an official topic in Module 3 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand Ladder diagram using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand ladder diagram should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand Ladder diagram means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Understand Ladder diagram [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

PLC programming

PLC programming is an official topic in Module 3 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify PLC programming using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, plc programming should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What PLC programming means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] PLC programming [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Case studies on PLC Ladder programming 3

Case studies on PLC Ladder programming 3 is an official topic in Module 3 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Case studies on PLC Ladder programming 3 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, case studies on plc ladder programming 3 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Case studies on PLC Ladder programming 3 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Case studies on PLC Ladder programming 3 □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Understand various Motion Controllers 3 elements, ladder diagram, PLC, programming of PLC- analog and digital I/Os, timers, counters, function blocks. Case studies on PLC ladder programming. Motion controllers- VFD, MLD, external relays and contactors. To understand telemetry system and Digital Communication channels in

Understand various Motion Controllers 3 elements, ladder diagram, PLC, programming of PLC- analog and digital I/Os, timers, counters, function blocks. Case studies on PLC ladder programming. Motion controllers- VFD, MLD, external relays and contactors. To understand telemetry system and Digital Communication channels in is an official topic in Module 3 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand various Motion Controllers 3 elements, ladder diagram, PLC, programming of PLC- analog and digital I/Os, timers, counters, function blocks. Case studies on PLC ladder programming. Motion controllers- VFD, MLD, external relays and contactors. To understand telemetry system and Digital Communication channels in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand various motion controllers 3 elements, ladder diagram, plc, programming of plc- analog and digital i/os, timers, counters, function blocks. case studies on plc ladder programming. motion controllers- vfd, mld, external relays and contactors. to understand telemetry system and digital communication channels in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Aspect	What to prepare
Definition/identity	What Understand various Motion Controllers 3 elements, ladder diagram, PLC, programming of PLC- analog and digital I/Os, timers, counters, function blocks. Case studies on PLC ladder programming. Motion controllers- VFD, MLD, external relays and contactors. To understand telemetry system and Digital Communication channels in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Understand various Motion Controllers 3 elements, ladder diagram, PLC, programming of PLC- analog and digital I/Os, timers, counters, function blocks. Case studies on PLC ladder programming. Motion controllers- VFD, MLD, external relays and contactors. To understand telemetry system and Digital Communication channels in [illegible] definition/meaning [illegible]. [illegible], steps, application [illegible]. Technical terms English-[illegible].

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Familiarize basic concept of PLC programming

  M3.01     Understand Sequence and logic control               1
Understanding


            Describe a programmable logic controller and
  M3.02                                                         2
Understanding
            its components

  M3.03     Understand Ladder diagram                           3
Applying

  M3.04     PLC programming                                     4
Applying

  M3.05     Case studies on PLC Ladder programming              3

  M3.06     Understand various Motion Controllers               3

Contents: Sequence control and programmable controllers – logic control and
sequencing
elements, ladder diagram, PLC, programming of PLC- analog and digital I/Os,
timers,
counters, function blocks. Case studies on PLC ladder programming. Motion
controllers-
VFD, MLD, external relays and contactors.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Process Control

This module is presented from the official syllabus scope for Industrial Automation Control. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Process Control
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Describe the different telemetry systems 3 Understanding Explain the concept of foundation Field bus

Describe the different telemetry systems 3 Understanding Explain the concept of foundation Field bus is an official topic in Module 4 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the different telemetry systems 3 Understanding Explain the concept of foundation Field bus using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the different telemetry systems 3 understanding explain the concept of foundation field bus should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the different telemetry systems 3 Understanding Explain the concept of foundation Field bus means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Describe the different telemetry systems 3 Understanding Explain the concept of foundation Field bus □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

6 and Profit bus DP Explain the concept of HART Communication

6 and Profit bus DP Explain the concept of HART Communication is an official topic in Module 4 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 and Profit bus DP Explain the concept of HART Communication using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 and profit bus dp explain the concept of hart communication should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 and Profit bus DP Explain the concept of HART Communication means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] 6 and Profit bus DP Explain the concept of HART Communication [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Applying Protocol System- Current Telemetry System - Motion balance andForce balance telemetry systems - Position Telemetry System- Digital communicationchannels-Fieldbus- advantagesoffieldbus-ProfibusDP-FunctionalelementsinFoundationFieldbus.Highway – AddressableRemoteTransducer-BenefitsofHARTfield communication protocol. Working of HART - Block diagram of HART digital communication System – HART Specification (Layers of HART). Text / Reference T/R Book Title/Author Industrial Instrumentation, Control and Automation, S. Mukhopadhyay, S. Sen and T1 A. K. Deb, Jaico Publishing House, 2013 R1 Jonathan Love, Process Automation Handbook, Springer 2007 R2 Curtis.D. Johnson, Process control instrumentation technology,PHI. R3 Liptak, Volume II, Instrument Engineers Hand book, Chilton Book Company. R4 Krishnakanth, Computer based Industrial Control, PHI. R5 Donald.P. Eckman, Automatic process control, WILLEY India. Online Resources Sl.No WebsiteLink 1 https://nptel.ac.in/courses/108105088 2 https://onlinecourses.nptel.ac.in/noc21_me67 /preview 3 https://scienceq.org/wp-content/uploads/2018/10/JAETV7I102.pdf 4 https://processcontrol.tv/tutorials 5 https://www.pidlab.com/en/virtual-labs

3 Applying Protocol System- Current Telemetry System - Motion balance andForce balance telemetry systems - Position Telemetry System- Digital communicationchannels-Fieldbus- advantagesoffieldbus-ProfibusDP-FunctionalelementsinFoundationFieldbus.Highway – AddressableRemoteTransducer-BenefitsofHARTfield communication protocol. Working of HART - Block diagram of HART digital communication System – HART Specification (Layers of HART). Text / Reference T/R Book Title/Author Industrial Instrumentation, Control and Automation, S. Mukhopadhyay, S. Sen and T1 A. K. Deb, Jaico Publishing House, 2013 R1 Jonathan Love, Process Automation Handbook, Springer 2007 R2 Curtis.D. Johnson, Process control instrumentation technology,PHI. R3 Liptak, Volume II, Instrument Engineers Hand book, Chilton Book Company. R4 Krishnakanth, Computer based Industrial Control, PHI. R5 Donald.P. Eckman, Automatic process control, WILLEY India. Online Resources Sl.No WebsiteLink 1 https://nptel.ac.in/courses/108105088 2 https://onlinecourses.nptel.ac.in/noc21_me67 /preview 3 https://scienceq.org/wp-content/uploads/2018/10/JAETV7I102.pdf 4 https://processcontrol.tv/tutorials 5 https://www.pidlab.com/en/virtual-labs is an official topic in Module 4 of Industrial Automation Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying Protocol System- Current Telemetry System - Motion balance andForce balance telemetry systems - Position Telemetry System- Digital communicationchannels-Fieldbus- advantagesoffieldbus-ProfibusDP-FunctionalelementsinFoundationFieldbus.Highway – AddressableRemoteTransducer-BenefitsofHARTfield communication protocol. Working of HART - Block diagram of HART digital communication System – HART Specification (Layers of HART). Text / Reference T/R Book Title/Author Industrial Instrumentation, Control and Automation, S. Mukhopadhyay, S. Sen and T1 A. K. Deb, Jaico Publishing House, 2013 R1 Jonathan Love, Process Automation Handbook, Springer 2007 R2 Curtis.D. Johnson, Process control instrumentation technology,PHI. R3 Liptak, Volume II, Instrument Engineers Hand book, Chilton Book Company. R4 Krishnakanth, Computer based Industrial Control, PHI. R5 Donald.P. Eckman, Automatic process control, WILLEY India. Online Resources Sl.No WebsiteLink 1 https://nptel.ac.in/courses/108105088 2 https://onlinecourses.nptel.ac.in/noc21_me67 /preview 3 https://scienceq.org/wp-content/uploads/2018/10/JAETV7I102.pdf 4 https://processcontrol.tv/tutorials 5 https://www.pidlab.com/en/virtual-labs using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying protocol system- current telemetry system - motion balance andforce balance telemetry systems - position telemetry system- digital communicationchannels-fieldbus- advantagesoffieldbus-profibusdp-functionalelementsinfoundationfieldbus.highway – addressableremotetransducer-benefitsofhartfield communication protocol. working of hart - block diagram of hart digital communication system – hart specification (layers of hart). text / reference t/r book title/author industrial instrumentation, control and automation, s. mukhopadhyay, s. sen and t1 a. k. deb, jaico publishing house, 2013 r1 jonathan love, process automation handbook, springer 2007 r2 curtis.d. johnson, process control instrumentation technology,phi. r3 liptak, volume ii, instrument engineers hand book, chilton book company. r4 krishnakanth, computer based industrial control, phi. r5 donald.p. eckman, automatic process control, willey india. online resources sl.no websitelink 1 https://nptel.ac.in/courses/108105088 2 https://onlinecourses.nptel.ac.in/noc21_me67 /preview 3 https://scienceq.org/wp-content/uploads/2018/10/jaetv7i102.pdf 4 https://processcontrol.tv/tutorials 5 https://www.pidlab.com/en/virtual-labs should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying Protocol System- Current Telemetry System - Motion balance andForce balance telemetry systems - Position Telemetry System- Digital communicationchannels-Fieldbus-advantagesoffieldbus-ProfibusDP-FunctionalelementsinFoundationFieldbus.Highway – AddressableRemoteTransducer-BenefitsofHARTfield communication protocol. Working of HART - Block diagram of HART digital communication System – HART Specification (Layers of HART). Text / Reference T/R Book Title/Author Industrial Instrumentation, Control and Automation, S. Mukhopadhyay, S. Sen and T1 A. K. Deb, Jaico Publishing House, 2013 R1 Jonathan Love, Process Automation Handbook, Springer 2007 R2 Curtis.D. Johnson, Process control instrumentation technology,PHI. R3 Liptak, Volume II, Instrument Engineers Hand book, Chilton Book Company. R4 Krishnakanth, Computer based Industrial Control, PHI. R5 Donald.P. Eckman, Automatic process control, WILLEY India. Online Resources Sl.No WebsiteLink 1 https://nptel.ac.in/courses/108105088 2 https://onlinecourses.nptel.ac.in/noc21_me67 /preview 3 https://scienceq.org/wp-content/uploads/2018/10/JAETV7I102.pdf 4 https://processcontrol.tv/tutorials 5 https://www.pidlab.com/en/virtual-labs means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 3 Applying Protocol System- Current Telemetry System - Motion balance andForce balance telemetry systems - Position Telemetry System- Digital communicationchannels-Fieldbus- advantagesoffieldbus-ProfibusDP-FunctionalelementsinFoundationFieldbus.Highway – AddressableRemoteTransducer-BenefitsofHARTfield communication protocol. Working of HART - Block diagram of HART digital communication System – HART Specification (Layers of HART). Text / Reference T/R Book Title/Author Industrial Instrumentation, Control and Automation, S. Mukhopadhyay, S. Sen and T1 A. K. Deb, Jaico Publishing House, 2013 R1 Jonathan Love, Process Automation Handbook, Springer 2007 R2 Curtis.D. Johnson, Process control instrumentation technology,PHI. R3 Liptak, Volume II, Instrument Engineers Hand book, Chilton Book Company. R4 Krishnakanth, Computer based Industrial Control, PHI. R5 Donald.P. Eckman, Automatic process control, WILLEY India. Online Resources Sl.No WebsiteLink 1 https://nptel.ac.in/courses/108105088 2 https://onlinecourses.nptel.ac.in/noc21_me67 /preview 3 https://scienceq.org/wp-content/uploads/2018/10/JAETV7I102.pdf 4 https://processcontrol.tv/tutorials 5 https://www.pidlab.com/en/virtual-labs □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Process Control

  M4.01     Describe the different telemetry systems            3
Understanding

            Explain the concept of foundation Field bus
  M4.02                                                         6
            and Profit bus DP
            Explain the concept of HART Communication
  M4.03                                                         3
Applying
            Protocol

            Series Test – II                                    1
Contents: Telemetry system- General Telemetry system blockDiagram- Voltage
Telemetry
System- Current Telemetry System - Motion balance andForce balance telemetry
systems -
Position     Telemetry       System-     Digital     communicationchannels-
Fieldbus-
advantagesoffieldbus-ProfibusDP-FunctionalelementsinFoundationFieldbus.Highway
–
 AddressableRemoteTransducer-BenefitsofHARTfield communication protocol. Working
of
 HART - Block diagram of HART digital communication System – HART Specification
 (Layers of HART).
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Define the concepts of process control.. (3 marks)

Answer: Begin with the standard definition or identity of *Define the concepts of process control.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Explain automatic control 3 Understanding Explain the process and controller. (3 marks)

Answer: Begin with the standard definition or identity of *Explain automatic control 3 Understanding Explain the process and controller*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 4 Understanding Characteristics. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding Characteristics*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Describe the different process control loops 5 Understanding Automatic control-Block diagram of process control - Error, set point, controlled variable, manipulated variable and measured variable-Elements of process control loop-Process characteristics - Process equation, Process Load, Process Lag and Self regulation, Control System Parameters - Error, Variable Range, Control Parameter Range, Control Lag –Dead Time, Cycling-Process Parameters-Temperature process control, Pressure Process control system-

Flow process control system, Level process control system. To understand the various control modes and to implement pneumatic. (3 marks)

Answer: Begin with the standard definition or identity of *Describe the different process control loops 5 Understanding Automatic control-Block diagram of process control - Error, set point, controlled variable, manipulated variable and measured variable-Elements of process control loop-Process characteristics - Process equation, Process Load, Process Lag and Self regulation, Control System Parameters - Error, Variable Range, Control Parameter Range, Control Lag –Dead Time, Cycling-Process Parameters-Temperature process control, Pressure Process control system-Flow process control system, Level process control system. To understand the various control modes and to implement pneumatic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain Describe different types of process 4 Understanding To understand the various control modes. (3 marks)

Answer: Begin with the standard definition or identity of *Describe different types of process 4 Understanding To understand the various control modes*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 6 Understanding continuous and discontinuous Understand the implementation of Electronic. (3 marks)

Answer: Begin with the standard definition or identity of *6 Understanding continuous and discontinuous Understand the implementation of Electronic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding Controllers. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding Controllers*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Illustrate Tuning of controllers 3 Applying neutralzone -multi position control Mode- Continuous Control Modes - proportional controlMode -proportional band and offset error - integral control Mode – reset rate – derivative control Mode – derivative time- composite control Modes- PI, PD and PID control Modes–Electronic Controllers-Electronic Error detector-Electronic P,PI,PD and PID controller using Op -amp-Tuning methods– open loop method-Ziegler Nichols method.. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate Tuning of controllers 3 Applying neutralzone -multi position control Mode- Continuous Control Modes - proportional controlMode -proportional band and offset error - integral control Mode – reset rate – derivative control Mode – derivative time- composite control Modes- PI, PD and PID control Modes–Electronic Controllers-Electronic Error detector-Electronic P,PI,PD and PID controller using Op -amp-Tuning methods– open loop method-Ziegler Nichols method.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Understand Sequence and logic control 1 Understanding Describe a programmable logic controller and. (3 marks)

Answer: Begin with the standard definition or identity of *Understand Sequence and logic control 1 Understanding Describe a programmable logic controller and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 2 Understanding its components. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding its components*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Understand Ladder diagram. (3 marks)

Answer: Begin with the standard definition or identity of *Understand Ladder diagram*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain PLC programming. (3 marks)

Answer: Begin with the standard definition or identity of *PLC programming*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Module 4

Define or explain Describe the different telemetry systems 3 Understanding Explain the concept of foundation Field bus. (3 marks)

Answer: Begin with the standard definition or identity of *Describe the different telemetry systems 3 Understanding Explain the concept of foundation Field bus*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 6 and Profit bus DP Explain the concept of HART Communication. (3 marks)

Answer: Begin with the standard definition or identity of *6 and Profit bus DP Explain the concept of HART Communication*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 3 Applying Protocol System- Current Telemetry System - Motion balance andForce balance telemetry systems - Position Telemetry System- Digital communicationchannels-Fieldbus- advantagesoffieldbus-ProfibusDP-FunctionalelementsinFoundationFieldbus.Highway – AddressableRemoteTransducer-BenefitsofHARTfield communication protocol. Working of HART - Block diagram of HART digital communication System – HART Specification (Layers of HART). Text / Reference T/R Book Title/Author

Industrial Instrumentation, Control and Automation, S. Mukhopadhyay, S. Sen and T1 A. K. Deb, Jaico Publishing House, 2013 R1 Jonathan Love, Process Automation Handbook, Springer 2007 R2 Curtis.D. Johnson, Process control instrumentation technology,PHI. R3 Liptak, Volume II, Instrument Engineers Hand book, Chilton Book Company. R4 Krishnakanth, Computer based Industrial Control, PHI. R5 Donald.P. Eckman, Automatic process control, WILLEY India. Online Resources Sl.No WebsiteLink 1 https://nptel.ac.in/courses/108105088 2 https://onlinecourses.nptel.ac.in/noc21_me67 /preview 3 https://scienceq.org/wp-content/uploads/2018/10/JAETV7I102.pdf 4 https://processcontrol.tv/tutorials 5 https://www.pidlab.com/en/virtual-labs. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying Protocol System- Current Telemetry System - Motion balance andForce balance telemetry systems - Position Telemetry System- Digital communicationchannels-Fieldbus- advantagesoffieldbus-ProfibusDP-FunctionalelementsinFoundationFieldbus.Highway – AddressableRemoteTransducer-BenefitsofHARTfield communication protocol. Working of HART - Block diagram of HART digital communication System – HART Specification (Layers of HART). Text / Reference T/R Book Title/Author Industrial Instrumentation, Control and Automation, S. Mukhopadhyay, S. Sen and T1 A. K. Deb, Jaico Publishing House, 2013 R1 Jonathan Love, Process Automation Handbook, Springer 2007 R2 Curtis.D. Johnson, Process control instrumentation technology,PHI. R3 Liptak, Volume II, Instrument Engineers Hand book, Chilton Book Company. R4 Krishnakanth, Computer based Industrial Control, PHI. R5 Donald.P. Eckman, Automatic process control, WILLEY India. Online Resources Sl.No WebsiteLink 1 https://nptel.ac.in/courses/108105088 2 https://onlinecourses.nptel.ac.in/noc21_me67 /preview 3 https://scienceq.org/wp-content/uploads/2018/10/JAETV7I102.pdf 4 https://processcontrol.tv/tutorials 5 https://www.pidlab.com/en/virtual-labs*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Define the concepts of process control.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Describe different types of process 4 Understanding To understand the various control modes**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Understand Sequence and logic control 1 Understanding Describe a programmable logic controller and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Describe the different telemetry systems 3 Understanding Explain the concept of foundation Field bus**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No WebsiteLink https://nptel.ac.in/courses/108105088
- https://onlinecourses.nptel.ac.in/noc21_me67 /preview
- https://scienceq.org/wp-content/uploads/2018/10/JAETV7I102.pdf https://processcontrol.tv/tutorials https://www.pidlab.com/en/virtual-labs

IN-PAGE SEARCH

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